Elrina Avionics Checklist

Don't forget to charge battery before launch day and make ALL UP TEST BEFORE BLACK POWDER

- **Verification before leaving base camp :**

Verify igniter wiring : **BOOSTER**	**Verify igniter wiring :** **SUSTAINER**
Follow wiring diagram to verify other connections	**Follow wiring diagram to verify other connections**
Type : [« Igniter name » : « Altimeter » : « Channel »]	Type : [« Igniter name » : « Altimeter » : « Channel »]
- Connection list :	- Connection list :
☐ Primary apogee : Blue Raven : Apogee	☐ Primary apogee : Blue Raven : Apogee
☐ Secondary apogee : Stratologger : Apogee	☐ Secondary apogee : Stratologger : Apogee
☐ Primary main : Blue Raven : Main	☐ Primary main : Blue Raven : Main
☐ Secondary main : Stratologger : Main	☐ Secondary main : Stratologger : Main
☐ Primary separation : Blue Raven : 3rd	☐ Primary 2nd stage ignition : Blue Raven : 3rd

Batteries verifications **BOOSTER**	
Take measure with voltmeter	
☐ 2S Li-ion Batteries 7.4V (BlueRaven) – min 7.4V	V
☐ 9 V alkaline Batterie (Stratologger)	V
☐ 2S Li-ion Battery Pack 7,4 V (ODB, IdéFIX, Cameras, RFD-900X, Featherweight GPS)	V

Batteries verifications **SUSTAINER**

Take measure with voltmeter

☐ 2S Li-ion Batteries 7,4V (BlueRaven) – min 7,4V ____ V

☐ First 9 V alkaline Batterie (Stratologger) ____ V

☐ 2S Li-ion Battery Pack 7,4 V (ODB, IdéFIX, Cameras, RFD-900X, Featherweight GPS) ____ V

Batteries verifications **GROUND STATION & PAD**

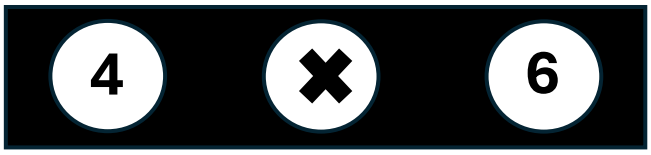
Take measure with voltmeter

☐ Check Battery pack “Ambatakum” is full (Ground Station)

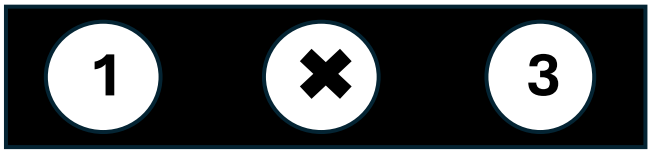
☐ Check Battery pack “Lilo” is full (Ground Station)

☐ Check Battery Pad Camera

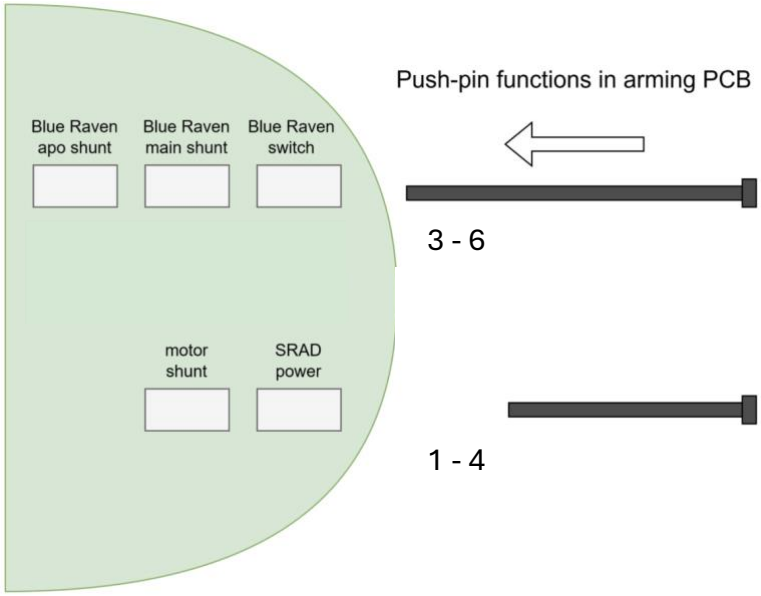
☐ Check Battery Featherweight Ground Station



SUSTAINER



BOOSTER



DON’T Forget material before leaving

All up test **WARNING ! MAKE BEFORE BLACK POWDER MATING**

SUSTAINER

☐ Active Statrologger with magnet

☐ Listen to Stratologger beeping **REFER BELOW**

☐ Verify programming

☐ Verify continuity

☐ Verify battery voltage (9,6V)

V

☐ Insert pushpin 6 **BEWARE OF MOTOR IGNIT**

☐ Verify BlueRaven startup

☐ Insert pushpin 4 halfway



☐ Verify Featherweight GPS startup

☐ Verify ODB/CAM/IDFX/RFD startup

☐ Insert pushpin 4 all the way in

☐ Verify shunt continuity

BOOSTER

☐ Insert pushpin 2

☐ Listen to Stratologger beeping **REFER BELOW**

☐ Verify programming

☐ Verify continuity

☐ Verify battery voltage (9,6V)

V

☐ Insert pushpin 2 - 3

☐ Verify BlueRaven startup

☐ Verify Featherweight GPS startup

☐ Verify ODB/CAM/IDFX/RFD startup

CELLPHONE

For booster

☐ Verify Featherweight (Booster).
FIX with Cellphone

☐ Verify continuity Featherweight (Booster).

☐ Verify Featherweight GPS (Booster) battery voltage (7.4 V)

V

For sustainer

☐ Verify Featherweight (Sustainer)
FIX with Cellphone.

☐ Verify continuity Featherweight (Booster).

☐ Verify Featherweight (Sustainer) battery voltage (7.4 V)

V

GROUND STATION

☐ Turn ground station on

☐ Turn IdéFIX Receiver on

☐ Verify IdéFIX is transmitting to receiver

☐ Verify ODB is transmitting to ground station

☐ Verify ODB data (GPS, altimeter, accelerometer)

☐ Verify ODB SD fix

STARTOLOGGER BEEP SEQUENCE

Digit	*Reported as:*
0	beep-beep-beep-beep-beep-beep-beep-beep-beep-beep
1	beep
2	beep-beep
3	beep-beep-beep
4	beep-beep-beep-beep
5	beep-beep-beep-beep-beep
6	beep-beep-beep-beep-beep-beep
7	beep-beep-beep-beep-beep-beep-beep
8	beep-beep-beep-beep-beep-beep-beep-beep
9	beep-beep-beep-beep-beep-beep-beep-beep-beep

Detected an anormal condition

Ok avec une croix si l'évènement n'arrive pas

Short siren + beep + siren	Total power loss during last flight	
Short siren + beep beep + siren	Momentary power loss during flight	
Short siren + beep beep beep+ siren	Drogue currrent exceeded 6 amps	
Short siren + beep beep beep beep + siren	Main current exceeded 6 amps	
Short siren + beep beep beep beep beep + siren	Problem detected with flash memory	
Short siren + beep beep beep beep beep beep + siren	Problem detected with pressure sensor DO NOT FLY	

Normal informations :

Ok avec croix si l'évènement arrive

Program set (7)	7x beep	
Digit main deploy altitude (1500)	1 x beep 5 x beep 1 x beep 1 x beep	
Ne pas tenir compte		
Check volatge battery (exemple 92 -> 9.2)	Voir bip 8.9 minimum	
Continuité dans tous les systèmes	Beep beep beep every 0,8 sec	

Ok pour stratologger ready for launch

DON'T Forget material before leaving

- **At Launch Site**

Bring laptop for Stratologger, Ground station for ODB, IdeFIX Receiver for IdeFix and cellphone for Featherweight on the launch pad, prepare sufficient battery.

Pre-launch pad (safety inspection)

☐ Wipe SD cards for cameras

☐ Wipe SD ODB

☐ Turn spot trace on

☐ Verify every connection (tirer sur chaque fil / verifier la solidité + wiring diagram)

☐ Check camera in good position

☐ Check Featherweight is in position “on” (switch)

☐ Mark connection with tape when verified

☐ Check wiring during assembly to prevent accident

Cheklist material for pad operation

☐ Ground station

☐ Laptop with Stratologger cable

☐ Security glasses

☐ Security gloves

☐ **Allen key**

☐ IdeFix receiver

☐ Cellphone with Featherweight app

☐ Table

☐ Tuck tape

☐ Pad camera

- **On the launch pad**

On the launch pad

SUSTAINER

☐ Active Stratologger with magnet

☐ Listen to Stratologger beeping **REFER BELOW**

☐ Verify programming

☐ Verify continuity

☐ Verify battery voltage (9,6V)

V

☐ Insert pushpin 6 **BEWARE OF MOTOR IGNIT**

☐ Verify BlueRaven startup



☐ Insert pushpin 4 halfway

☐ Verify Featherweight GPS startup

☐ Verify ODB/CAM/IDFX/RFD startup

☐ Insert pushpin 4 all the way in

☐ Verify shunt continuity

BOOSTER

☐ Insert pushpin __2_

☐ Listen to Stratologger beeping **REFER BELOW**

☐ Verify programming

☐ Verify continuity

☐ Verify battery voltage (9,6V)

V

☐ Insert pushpin __2 - 3_

☐ Verify BlueRaven startup

☐ Verify Featherweight GPS startup

☐ Verify ODB/CAM/IDFX/RFD startup

GROUND STATION

☐ Turn ground station on

☐ Turn IdéFIX Receiver on

☐ Verify IdéFIX is transmitting to receiver

☐ Verify ODB is transmitting to ground station

☐ Verify ODB data (GPS, altimeter, accelerometer)

☐ Verify ODB SD fix

CELLPHONE

For booster

☐ Verify Featherweight (Booster).
FIX with Cellphone

☐ Verify continuity Featherweight (Booster).

☐ Verify Featherweight GPS (Booster) battery voltage (7.4 V)

V

For sustainer

☐ Verify Featherweight (Sustainer)
FIX with Cellphone.

☐ Verify continuity Featherweight (Booster).

☐ Verify Featherweight (Sustainer) battery voltage (7.4 V)

V

Digit	*Reported as:*
0	beep-beep-beep-beep-beep-beep-beep-beep-beep-beep
1	beep
2	beep-beep
3	beep-beep-beep
4	beep-beep-beep-beep
5	beep-beep-beep-beep-beep
6	beep-beep-beep-beep-beep-beep
7	beep-beep-beep-beep-beep-beep-beep
8	beep-beep-beep-beep-beep-beep-beep-beep
9	beep-beep-beep-beep-beep-beep-beep-beep-beep

Detected an anormal condition

Ok avec une croix si l'évènement n'arrive pas

Short siren + beep + siren	Total power loss during last flight	
Short siren + beep beep + siren	Momentary power loss during flight	
Short siren + beep beep beep+ siren	Drogue currrent exceeded 6 amps	
Short siren + beep beep beep beep + siren	Main current exceeded 6 amps	
Short siren + beep beep beep beep beep + siren	Problem detected with flash memory	
Short siren + beep beep beep beep beep beep + siren	Problem detected with pressure sensor DO NOT FLY	

Normal informations :

Ok avec croix si l'évènement arrive

Program set (7)	7x beep	
Digit main deploy altitude (1500)	1 x beep 5 x beep 1 x beep 1 x beep	
Ne pas tenir compte		
Check volatge battery (exemple 92 -> 9.2)	Voir bip 8.9 minimum	
Continuité dans tous les systèmes	Beep beep beep every 0,8 sec	

Ok pour stratologger ready for launch

On the launch pad

CELLPHONE FINAL VERIF

BOOSTER

Blue Raven

☐ Verify continuity of each channel Battery (7.4) _____ V

☐ Verify flight program

Stratologger

☐ Verify continuity of each channel Battery (9.6) _____ V

☐ Verify flight program

Featherweight GPS

☐ Verify fix Ground station

☐ Verify fix cell phone Battery (7.4) _____ V

☐ Verify frequency

SUSTAINER

Blue Raven

☐ Verify continuity of each channel Battery (7.4) _____ V

☐ Verify flight program

Stratologger

☐ Verify continuity of each channel Battery (9.6) _____ V

☐ Verify flight program

Featherweight GPS

☐ Verify fix Ground station

☐ Verify fix cell phone Battery (7.4) _____ V

☐ Verify frequency

All systems are go for launch!

☐ **Get away from pad**